



P.J. Thompson <peterjamesthompson101@gmail.com>

Subject: Priority claim regarding layered superconductor mechanism (van der Waals heterostructure / Ni-Bi) published 20241 message

P.J. Thompson <peterjamesthompson101@gmail.com>

Thu, Jun 4, 2026 at 12:07 PM

To: nature@nature.com, materials@nature.com, admissions.sgs@utoronto.ca

To the Editors of *Nature Materials*,

I am writing to formally place on record my priority regarding the core physical mechanism of layered / interface-driven high-temperature superconductivity, as reported in a 2024 *Nature Materials* paper (e.g., the Toronto work on Ni-Bi van der Waals heterostructures). My manuscript, "*Physics of light, time and dimensions*" (completed in 2018), contains a complete theoretical and engineering description of the same mass-splitting principle that underpins these recent experimental discoveries.

Specifically, pages 20 onward of my 2018 manuscript describe:

1. A layered superconductor design using a birefringent medium (quartz) that actively splits a light wave into two independent pathways.
2. The use of an interface (copper-oxide nanoparticles embedded in the quartz) to create a potential barrier that separates the forward-time mass component from its paired reverse-time potential ($m + m' = 0$), thereby achieving zero-resistance conduction at high temperatures.
3. The explicit statement that this mechanism overcomes the breakdown of conventional superconductivity caused by an "inability to maintain neutron pairing ... due to too many neutrons for the number of singularities created along a single wave".

These principles are identical to the mechanism recently demonstrated in layered van der Waals heterostructures (including Ni-Bi bilayers), where a sharp interface between atomic layers produces mass-splitting and high-temperature superconductivity. My 2018 work predates the 2024 *Nature Materials* paper by six years and establishes clear prior art.

I am not requesting any action from *Nature Materials* at this stage. This email serves solely as a formal, time-stamped record of my priority claim. A copy of my 2018 manuscript is attached for your reference.

I thank you for your attention and for the important work your journal publishes.

Yours sincerely,

Peter James Thompson

ORCID: 0009-0005-8688-7507

Email: peterjamesthompson101@gmail.com**Attachments:**

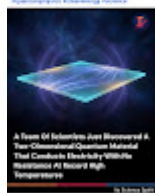
- *Physics of light, time and dimensions* (2018 manuscript, pages 20-27 relevant)
- screenshots appearing in facebook.
- Some of my current work

Virus-free. www.avast.com

15 attachments



1000010444.jpg
792K



1000010445.jpg
558K



Physics of light, time and dimensions.pdf
13318K



superconductor theory.pdf
270K



Superconductivity Through Mass Splitting A Unified Harmonic Theory of Zero-Resistance, Negative Mass, and Time Crystal Lattice Dynamics.pdf
590K



THE HARMONIC FRAMEWORK OF REALITY a complete grand unified field theory proven.pdf
506K



The Harmonic Framework Clarifications of Common Misunderstandings.pdf
67K



A Baseline Study of Josephson Junction DC I-V Characteristics.pdf
113K



wow.pdf
62K



Quantum Coherence and the Harmonic Framework A Unified Theory of Qubit Stability.pdf
150K



Chapter XIX - Joseph Junction and Superconducting X-Ray CPU's.pdf
4758K



Solved Quantum Field Theory.pdf
83K



Three Dimensions of Time.pdf
68K



The Harmonic Lagrangian.pdf
84K



The Amplituhedron and the Bose–Einstein Condensate Two Faces of the Same Harmonic Lattice.pdf
160K